Lecture 6: Math Agents

DSAA 6000J Foundations of Language Agents
2026 Spring

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DATA SCIENCE AND ANALYTICS THRUST
信息枢纽
INFORMATION HUB

AI agents in expert domains

- Finance
- Medicine
- Mathematics
- etc.,

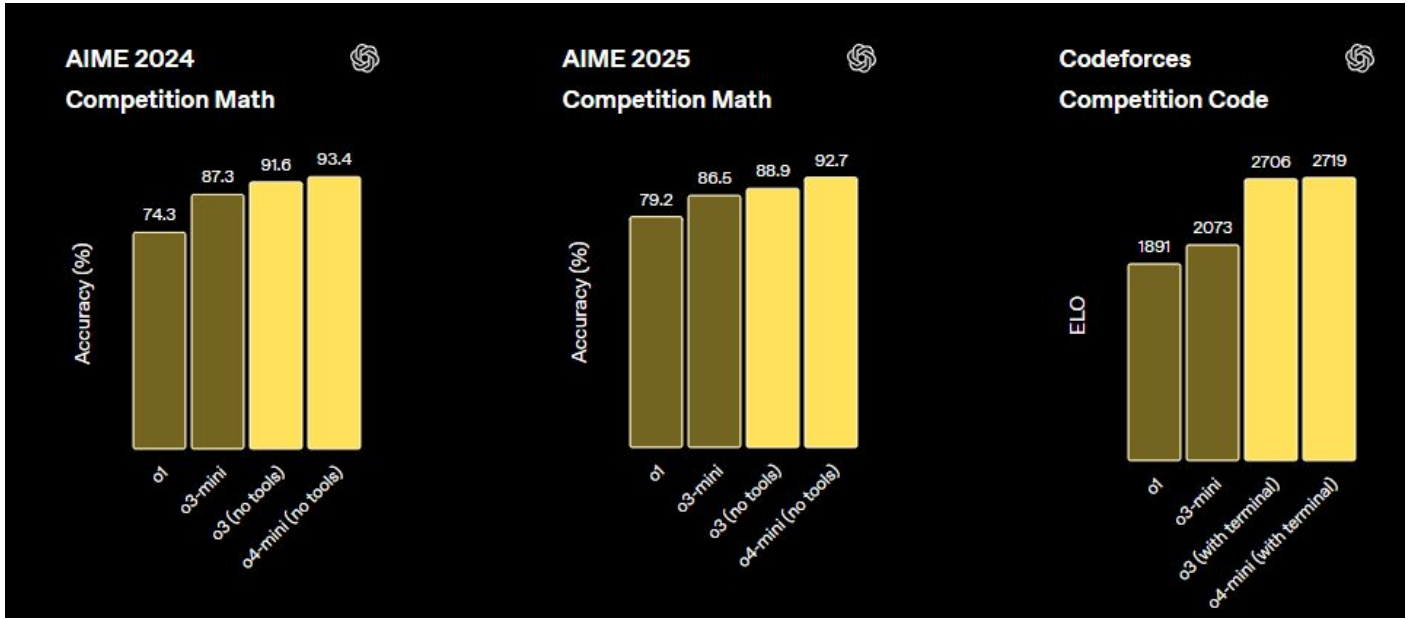
AI agents in expert domains

- Mathematics
  - Open-ended dialogue
  - Come up with counterexamples
  - Help write proofs
  - ...

Mathematics, a root node to intelligence

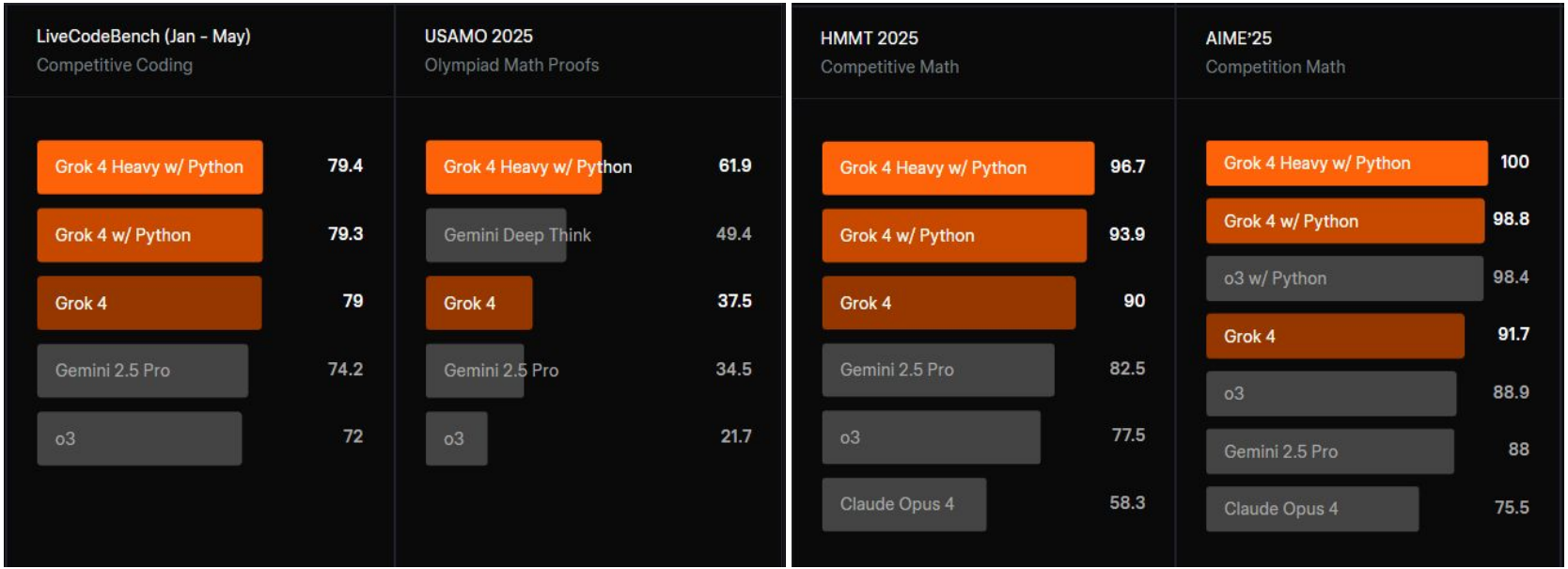
- *We have used maths to both describe, predict and shape the natural world.* – Thomas HUBERT
- A root node to intelligence
  - Reasoning & Planning
  - Generalisation & Abstraction
  - Knowledge & Creativity
  - Open ended & Unbounded complexity

AI Arms Race in Math and Coding



Introducing OpenAI o3 and o4-mini. OpenAI 2025

AI Arms Race in Math and Coding



Grok 4. XAI 2025

AI Arms Race in Math and Coding

AI|MO
ARTIFICIAL INTELLIGENCE MATHEMATICAL OLYMPIAD

Home About Participate Team Updates Contact

Update: The Third Progress Prize has launched

$10mn AI Mathematical Olympiad Prize Launches

XTX Markets has launched a new $10mn challenge fund, the **Artificial Intelligence Mathematical Olympiad Prize** (AIMO Prize).

The fund intends to spur the open development of AI models that can reason mathematically, leading to the creation of a publicly-shared AI model capable of winning a gold medal in the International Mathematical Olympiad (IMO).

There was a presentation of progress at the 65th IMO, held in

Bath, England in July 2024.

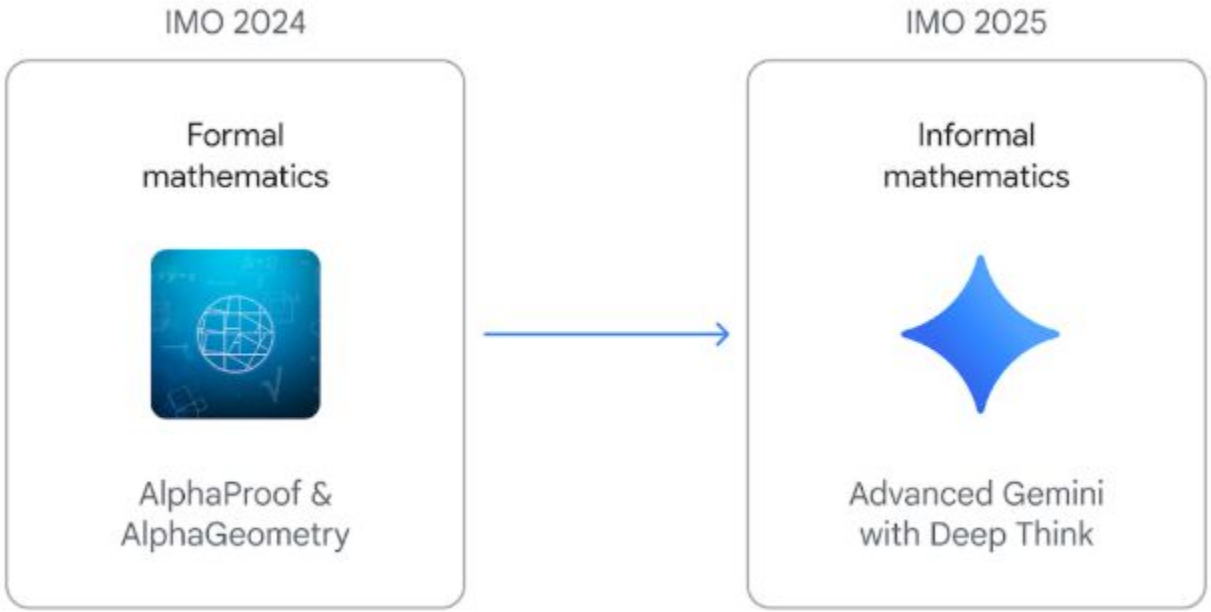
The AIMO Prize [..] will help compare different AI problem solving strategies at a technical level, in a manner that will be accessible and appealing to the broader public.

Terence Tao, UCLA

I am sure that many people will be following the AIMO Prize with great excitement, to see when, in the future, AI will match the world's brightest young minds.

Gregor Dolinar, IMO President

AI Arms Race in Math and Coding



Gemini with Deep Think officially achieves gold-medal standard at the IMO. Google 2025

AI Arms Race in Math and Coding

nature

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NEWS | 04 December 2025

DeepSeek's self-correcting AI model aces tough maths proofs

The mathematical reasoning model performed as well as humans at prestigious international mathematics competitions.

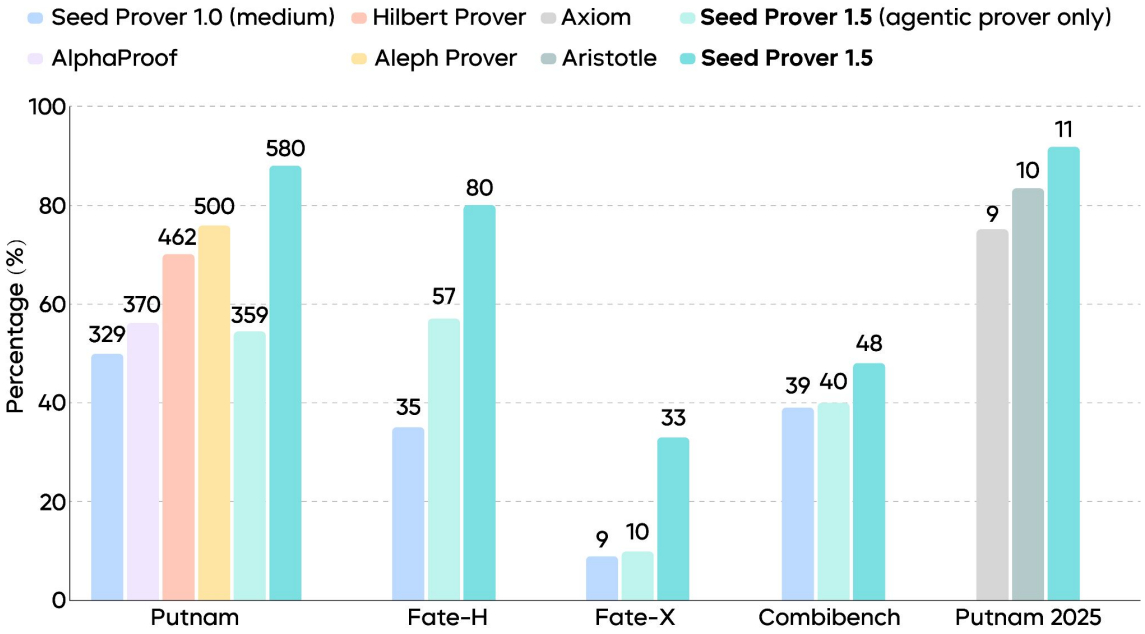
The model, DeepSeekMath-V2, scored 118 out of 120 points on questions from the 2024 William Lowell Putnam Mathematical Competition, beating the top human score of 90. The model also performed at the level of gold-medal winners in the International Mathematical Olympiad (IMO) 2025 and the 2024 China Mathematical Olympiad. The results are described in a preprint[1] posted on arXiv on 27 November.

"We are at a point where AI is about as good at maths as a smart undergraduate student," says Kevin Buzzard, a mathematician at Imperial College London. "It is very exciting."

In February, AlphaGeometry 2, an AI problem solver created by Google DeepMind in London, also achieved a gold-level performance in the IMO. The feat was repeated in July by Gemini's Deep Think, which is owned by DeepMind.

DeepSeekMath-V2: Towards Self-Verifiable Mathematical Reasoning. arXiv 2025

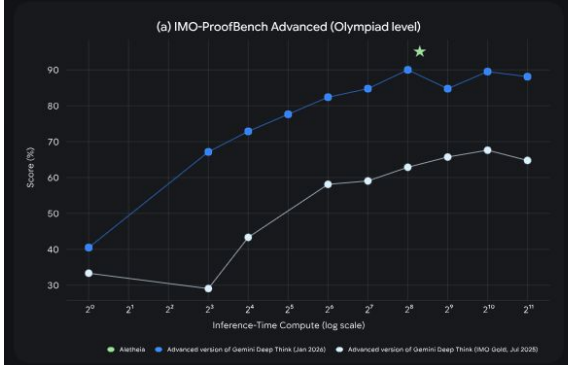
AI Arms Race in Math and Coding

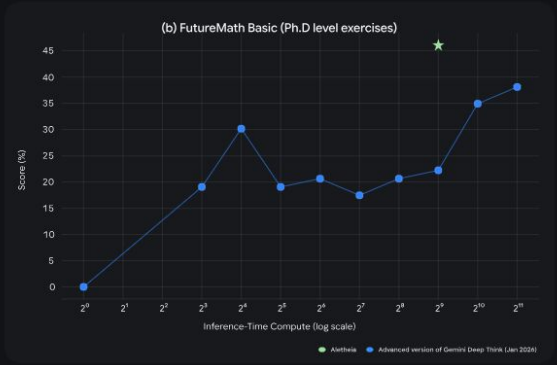


Seed Prover 1.5: Advanced Mathematical Reasoning through a Novel Agentic Architecture. ByteDance Seed 2025

AI Arms Race in Math and Coding

Since achieving IMO Gold-medal standard in July 2025, Gemini Deep Think has progressed rapidly, scoring up to 90% on the IMO-ProofBench Advanced test as inference-time compute scales. We demonstrated that the scaling law continues to hold as we progress beyond Olympiad level into PhD-level exercises (per our internal FutureMath Basic benchmark). Notably, Aletheia demonstrated that higher reasoning quality can be achieved at a lower inference-time compute.





The latest advanced version of Deep Think, as of Jan 2026, has significantly outperformed the IMO-Gold version (Jul 2025) on Olympiad-level problems. Aletheia makes further leaps in terms of reasoning quality with lower inference-time compute. All results were graded by human experts.

The inference-time scaling law also transfers to PhD-level exercises.

Accelerating Mathematical and Scientific Discovery with Gemini Deep Think. Google DeepMind 2026

AI Arms Race in Math and Coding

For research-level math, Aletheia has already enabled several advancements, produced via varying levels of autonomous research:

- Reliable autonomous research. A research paper (Feng26) generated by AI without any human intervention, which calculates certain structure constants in arithmetic geometry called eigenweights.
- AI-guided collaboration. A research paper (LeeSeo26) demonstrating human-AI collaboration in proving bounds on systems of interacting particles called independent sets.
- An extensive semi-autonomous evaluation (Feng et al., 2026b) of 700 open problems on Bloom's Erdős Conjectures database, including autonomous solutions to four open questions listed there. On Erdős-1051, our model autonomously solved and helped lead to a generalization reported in a research paper (BKKKZ26).

The agent also contributed intermediate propositions on two further papers, (FYZ26) and (ACGKMP26). It is also of note that there has been prior work using Gemini for research-level math at a smaller scale in terms of collaborations and the number of problems tackled.

Accelerating Mathematical and Scientific Discovery with Gemini Deep Think. Google DeepMind 2026

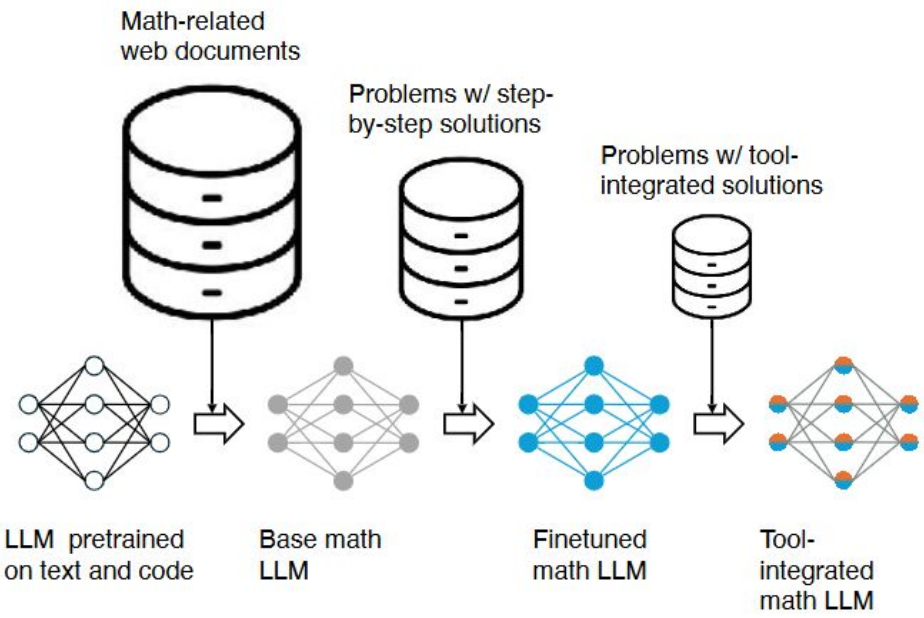
Why Math and Coding?

- Proxies for complex reasoning and planning
  - Important in human intelligence; challenging for LLMs
  - Unlimited applications: travel planning, calendar scheduling, etc.
- Relatively easy to evaluate
  - Math: check the answers
  - Coding: run unit tests
  - Writing a crime fiction? Composing a symphony?

How LLMs are Trained to Solve MathProblems?

- **Supervised finetuning (SFT):** “Good data is all you need!”
- **Reinforcement learning (RL):** “Verifiability is all you need!”
- Methods are straightforward, but the devil is in the details, e.g., data curation/cleaning, infrastructures for training and inference

Supervised Finetuning on Mathematical Data



Problem: Suppose that the sum of the squares of two complex numbers x and y is 7, and the sum of their cubes is 10. List all possible values for $x + y$, separated by commas.

Solution: Let's use `sympy` to calculate and print all possible values for $x + y$.

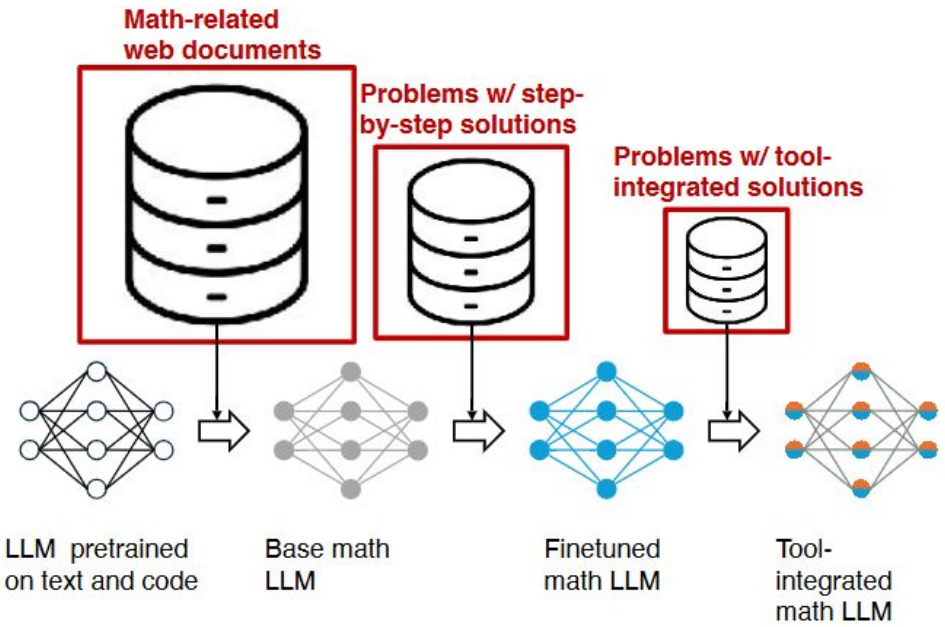
```
def possible_values():
    x, y = symbols("x y")
    eq1 = Eq(x**2 + y**2, 7)
    eq2 = Eq(x**3 + y**3, 10)
    solutions = solve((eq1, eq2), (x, y))
    return [simplify(sol[0] + sol[1]) for sol in solutions]

print(possible_values())
```

```
>>> [-5, -5, 1, 1, 4, 4]
```

Removing duplicates, the possible values for $x + y$ are \boxed{-5, 1, 4}

Formal Reasoning Meets LLMs: Towards AI for Mathematics andVerification. Meta 2025

Supervised Finetuning on Mathematical Data



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Formal Reasoning Meets LLMs: Towards AI for Mathematics andVerification. Meta 2025

Supervised Finetuning on Mathematical Data

- Training data is foremost important
  - Problems + (step-by-step, tool-integrated)solutions curated by humans and LLMs
  - Size of largest public datasets: ~900K [NuminaMath-1.5]

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Supervised Finetuning on Mathematical Data

- *What if the data has final answers but not intermediate steps ?*

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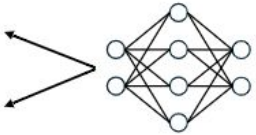
Reinforcement Learning on VerifiableProblems

✓ Solution: ... \boxed{-5, 1, 4}

✗ Solution: …\boxed{-2, 6}



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- Verify the model's solution by comparing the **final answer** with the ground truth

Reinforcement Learning on VerifiableProblems

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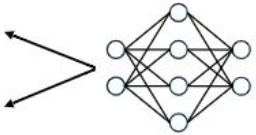

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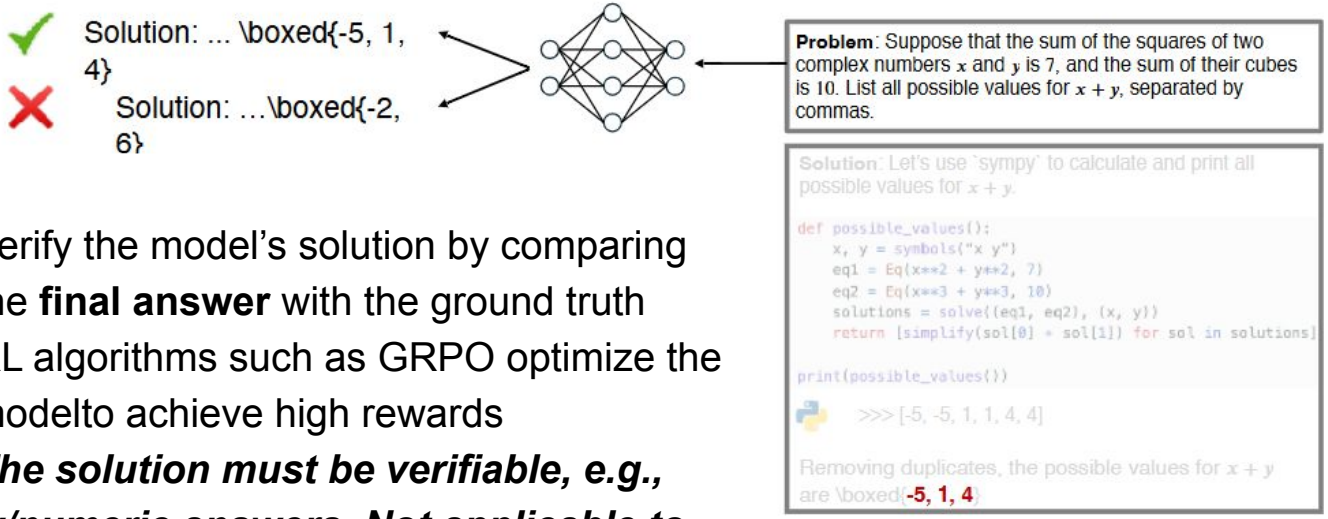
- Verify the model's solution by comparing the **final answer** with the ground truth
- RL algorithms such as GRPO optimize the modelto achieve high rewards
  - Popularized by DeepSeek-R1

Reinforcement Learning on VerifiableProblems



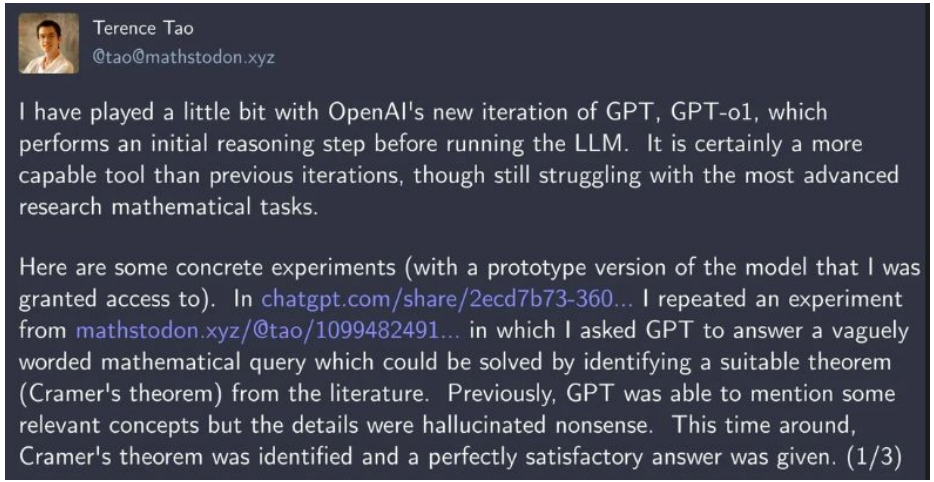
- Verify the model's solution by comparing the **final answer** with the ground truth
- RL algorithms such as GRPO optimize the modelto achieve high rewards
- ***The solution must be verifiable, e.g., w/numeric answers. Not applicable to proofs?***

How LLMs are Trained to Solve MathProblems?

- State-of-the-art math LLM ≈ strong pretrained model + two post-trainingtechniques + marvelous engineering
  - Supervised finetuning (SFT): “**Good data** is all you need!”
  - Reinforcement learning (RL): “**Verifiability** is all you need!”

Gap 1: Pre-college Math -> Advanced Math

- Existing successes are mostly on pre-college math, e.g., AIME, IMO
- LLMs struggle with more advanced math, e.g., mathematical research



Terence Tao
@tao@mathstodon.xyz

I have played a little bit with OpenAI's new iteration of GPT, GPT-o1, which performs an initial reasoning step before running the LLM. It is certainly a more capable tool than previous iterations, though still struggling with the most advanced research mathematical tasks.

Here are some concrete experiments (with a prototype version of the model that I was granted access to). In chatgpt.com/share/2ecd7b73-360... I repeated an experiment from mathstodon.xyz/@tao/1099482491... in which I asked GPT to answer a vaguely worded mathematical query which could be solved by identifying a suitable theorem (Cramer's theorem) from the literature. Previously, GPT was able to mention some relevant concepts but the details were hallucinated nonsense. This time around, Cramer's theorem was identified and a perfectly satisfactory answer was given. (1/3)

Gap 2: Guessing Answers -> Writing Proofs

- LLMs struggle to generate valid proofs

William Lowell PUTNAM Mathematical Competition

to be quite a big one. For example I was very unimpressed by the ChatGPT answers to the recent Putnam exam posted here — as far as I can see only question B4 was answered adequately by the machine, most other answers are worth one or two out of 10 at most. So I was expecting this dataset to remain pretty unattackable for a couple of years.

Can AI do maths yet? Thoughts from a mathematician. arXiv 2024

Gap 2: Guessing Answers -> Writing Proofs

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MAA MATHEMATICAL ASSOCIATION OF AMERICA

tors, we evaluated several state-of-the-art reasoning models on the six problems from the 2025 USAMO within hours of their release. Our results reveal that all tested models struggled significantly, achieving less than 5% on average. Through

Proof or Bluff? Evaluating LLMs on 2025 USA Math Olympiad. arXiv 2024

LLMs Alone are Not Enough

- Current math LLMs rely heavily on data and verifiability
- Data scarcity
  - Limited to data-rich domains, e.g., pre-college math
  - Cannot tackle advanced math or proofs
- Lack of verifiability
  - Solutions can only be evaluated by comparing with the ground truth•
  - Limited to problems with numeric solutions, e.g., GSM8K, MATH
  - Not applicable to most problems in advanced math

Formal Mathematical Reasoning

- Mathematical reasoning grounded in formal systems, e.g.,
  - First/higher-order logic
  - Dependent type theory
  - Computer programs & formal specifications•
- Formal environments can verify proofs and provide automatic feedback
  - Verification enables rigorous evaluation of reasoning
  - Learning from feedback mitigates data scarcity•
- Integrating formal reasoning and LLMs' informal reasoning

Formal Mathematical Reasoning: A New Frontier in AI. ICML 2025

A Brief History of Formalisation in Mathematics

- Proofs & Axiomatisation
  - Importance of proofs discovered by the ancient Greeks
  - Since then, what constitutes a mathematical proof has evolved.

A Brief History of Formalisation in Mathematics

- From words to symbols

A square and ten times its root are equal to thirty-nine.

$$x^2 + 10x = 39$$

Take the coefficient of the root, divide it into two parts, and multiply one of them by itself.
Add this to thirty-nine
Take the square root of sixty-four which is eight
Subtract from this one of the halves of the coefficient of the root.
The result is the value of the root which is three.

$$-\frac{b}{2} \pm \sqrt{\frac{b}{2}^2 - c}$$

Informal mathematics

- Math as raw data (text, images, ...)
  - Flexible
  - Widely used
  - Difficult to check

Input:
Let r be a root of $x^2 + 5x + 7 = 0$. Compute $(r-1)(r+2)(r+6)(r+3)$.

LLEMMA 34B solution:
We have

$$\begin{aligned}(r-1)(r+2)(r+6)(r+3) &= (r^2+r-2)(r^2+9r+18)\\ &= r^4+10r^3+25r^2+19r-36\\ &= r^4+5r^3+7r^2+5r+7+5r^3+20r^2+14r-36\\ &= r^4+5r^3+25r^2+19r-29\\ &= (r^2+5r+7)(r^2+4r-5)\\ &= (0)(r^2+4r-5)\\ &= \boxed{0}.\end{aligned}$$

Final Answer: The final answer is 0.

Language model solution.

Formal mathematics

- Math as source code
  - Write a specification (e.g., 1+1=2)
  - Write a proof
  - Automatically checked
    - Code compiles ≡ correct proof

$1 + 1 = 2$

proof ✓

```
lemma one_plus_one_equals_two:
  shows "1 + 1 = 2"
proof -
  have "1 + 1 = Suc (0 + 1)" by simp
  also have "... = Suc 1" by simp
  also have "... = 2" by simp
  finally show ?thesis by simp
qed
```

Math as source code.

Formal mathematics

- Math as source code
  - Write a specification (e.g., 1+1=2)
  - Write a proof
  - Automatically checked
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Lean Isabelle Coq

Theorem proving languages

How is formal math used in practice?

- Growing use in mathematics
- Lean Mathlib project: 1+ million lines of code, 300+ contributors

Terence Tao
@tao@mathstodon.xyz

Finished formalizing in #Lean4 the proof of an actual new theorem (Theorem 1.3) in my recent paper arxiv.org/abs/2310.05328 :

Terence Tao's Lean formalization project (October 2023)

How is formal math used in practice?

- Collaboration
  - Break down a big problem into multiple pieces
  - Anyone can submit code to solve a piece
  - We know we can trust the code since it is automatically checked!

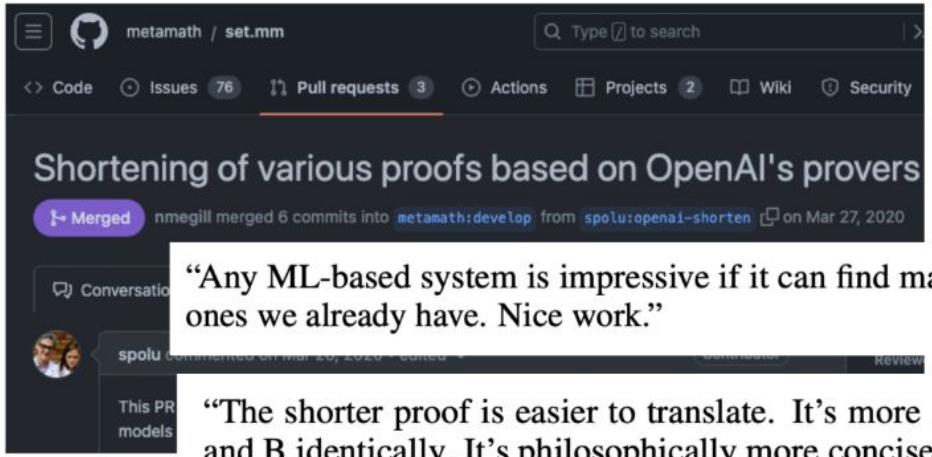
How is formal math used in practice?

- Collaboration
  - Break down a big problem into multiple pieces
  - Anyone can submit code to solve a piece
  - We know we can trust the code since it is automatically checked!
- Instant feedback
- Guaranteed correctness

Why is AI + formal math important?

- Verifiable
  - Prevent incorrect math and code generation
  - Feedback signal for learning•
- Tests reasoning
  - From easy: 1+1 = 2
  - To hard: Fermat’s Last Theorem

Early Effort: GPT-*f*



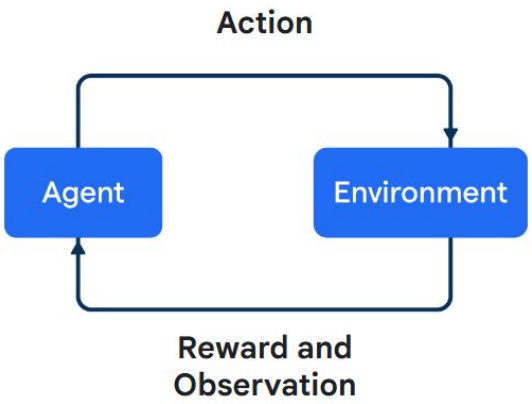
"Any ML-based system is impressive if it can find many shorter proofs than the ones we already have. Nice work."

"The shorter proof is easier to translate. It's more symmetric in that it treats A and B identically. It's philosophically more concise in that it doesn't rely on the existence of a universal class of all sets."

Generative Language Modeling for Automated Theorem Proving. OpenAI 2020

Recap: What's RL

- RL is trial and error learning
- An agent learns by interacting with anenvironment to maximize reward.



Recap: RL is A Proven Recipe to Superintelligence

Atari DQN

AlphaGo

AlphaGoZero

AlphaZero

AlphaStar

MuZero

Stratego

AlphaDev

AlphaTensor

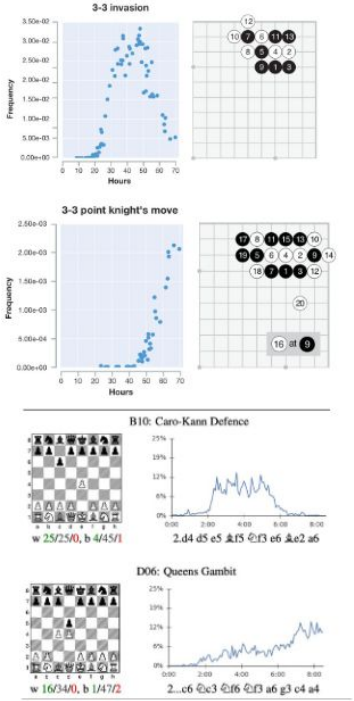
Fusion

The Zero Philosophy

- If an agent can learn to master an environment tabula rasa, then you demonstrably have a system that is discovering and learning new knowledge by itself.
- It also means that this algorithm is general and should apply to other domains.
- This is particularly important to advance science.

AlphaGoZero & AlphaZero

- Learned simply by playing games against itself, starting from completely random play.
- Rediscovered human patterns, accumulating thousands of years of human knowledge in a matter of days.
- Ultimately discarded some in preference of new discoveries that humans don't even know about.



What made those systems Superhuman?

- Scaled up trial and error
- Grounded feedback signal
- Search
- Curriculum

What made those systems Superhuman?

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AlphaZero is the agent.

+

Lean is the environment.

When RL meets Formal Maths

AlphaZero is the agent.

Lean is the environment.

Olympiad-level formal mathematical reasoning with reinforcement learning. Nature 2020

AlphaProof: Master Plan

- Lean gives us a way to scale up trial and error with
  - An environment to explore mathematics completely in silico
  - A perfect feedback signal for proving
- We can therefore reach superhuman intelligence and discover new truths, just like in Go/chess/Tensor decomposition... provided:
  - We can generate high enough quality + quantity problems
  - RL works

From Mathematical Reasoning to Domain Intelligence in LLMs: Recipes, Reliability, and Open Questions for Real-World AI

- **Abstract:** *LLMs have made remarkable progress in mathematical reasoning, where abundant data, clear benchmarks, and favorable scaling properties have enabled rapid advances. In this talk, I will present several of our recent works in this area, including the winning recipe behind our AIMO Prize 2 solution, a finding on how reflection emerges in System 2 reasoning, and a formal verification approach for improving the reliability of mathematical reasoning.I will then briefly turn to a broader perspective: although mathematical reasoning has become a highly visible and fast-moving direction, real-world domain applications remain much harder. I will close by sharing several open questions on domain intelligence that may shape the next phase of large language model research.*
- **Date:** 18th March

From Mathematical Reasoning to Domain Intelligence in LLMs: Recipes, Reliability, and Open Questions for Real-World AI

- Dr. Yan Xu is a Senior Researcher at Tencent, where she works on Financial LLMs. Her research focuses on NLP, LLMs and reasoning. Previously, her work centered on enhancing the mathematical capabilities of LLMs through large-scale pre-training data construction, instruction tuning, as well as formal verification-based reasoning methods. Yan received her Ph.D. from the HKUST, advised by Prof. Pascale Fung. Her research has been published in ACL, ICML, EMNLP, and ACM Computing Surveys. Her work has received several recognitions, including the **Gold Medal in the Kaggle AI Mathematical Olympiad** – Progress Prize 2, Best Student Paper Award at the ACL DialDoc Workshop, and Area Chair Award at AACL 2023.

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Project Discussion

- How to improve your project with math reasoning techniques？
- **Project Proposal Deadline: March 20th, 11:59 PM**